

```
---  
title: "Markdown Formatting & Publication Guide"  
date: "2026-06-19"  
---
```

```
## Welcome
```

Markdown Formatting & Publication Guide

The definitive reference for writing, styling, and debugging content on the Dr. Neal Aggarwal Flask engine.

The Architecture of a Post

Every post on the platform is processed by a bespoke Flask engine using the Python Markdown library. The parser relies on strict syntax rules and custom CSS injections to convert your raw text into the site's signature dark technical theme.



The Strict Filename Rule

The Flask app derives the post URL directly from the filename.
Getting this wrong breaks the routing.

YYYY-MM-DD-slug-words-separated-by-hyphens.md



**new_post on neural
networks.md**

Broken URL

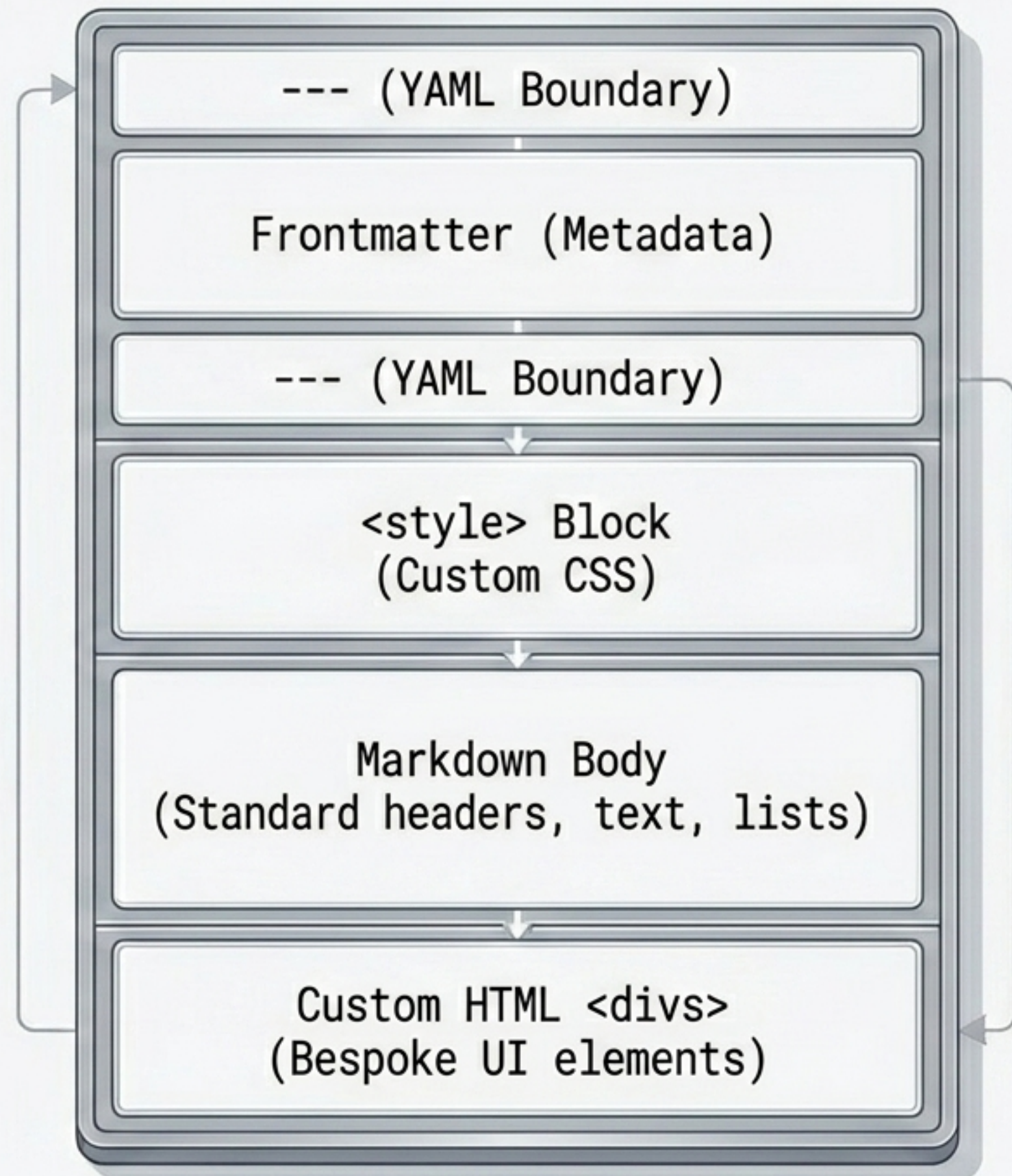


**2026-06-19-how-deep-neural-
networks-really-work.md**

Valid URL

The Anatomy of a File

Browsers and parsers render top-to-bottom.
The structure must strictly follow this sequential order
to prevent CSS failures or broken metadata.



YAML Frontmatter

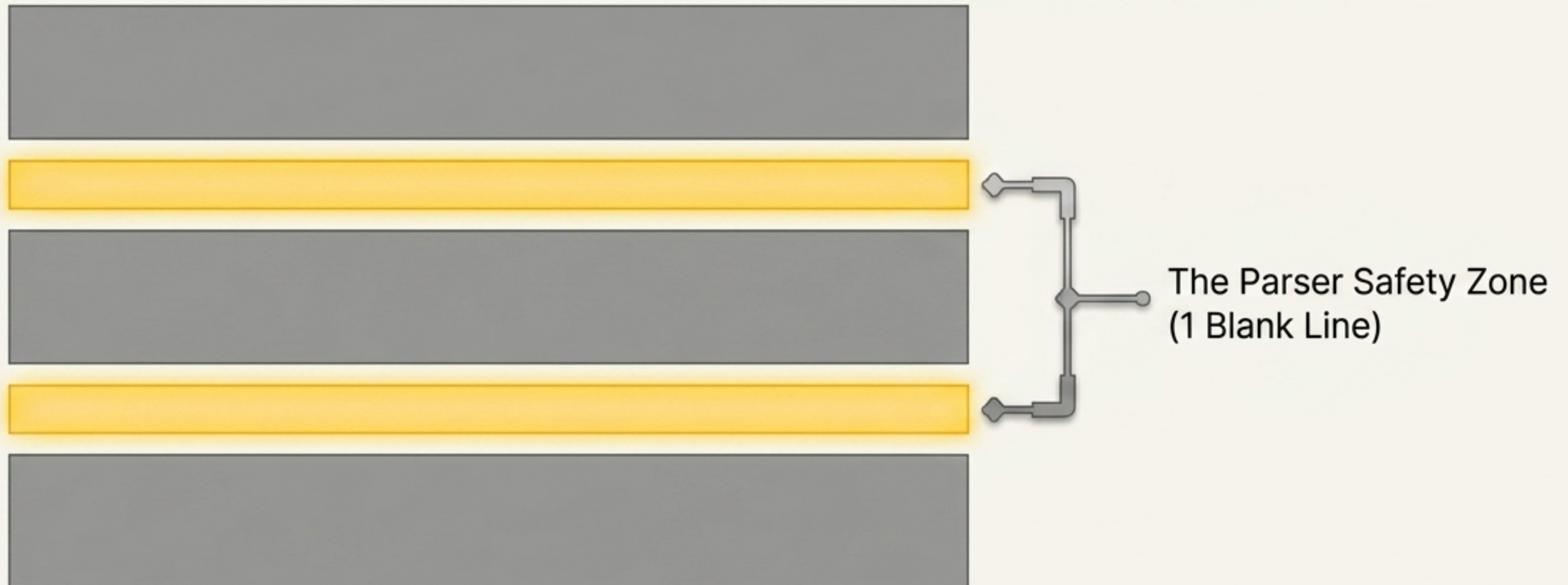
Rule Alert: This block must be the absolute first thing in the file. No blank lines before the opening ---.

```
---
title: Deep Neural Networks
date: 2026-06-19
category: AI Engineering
tags: [ml, backend]
summary: A technical deep dive.
level: Advanced
read_time: 15 min
---
```

Required Keys	Optional Keys
• title	• level
• date	• read_time
• category	• featured
• tags	
• summary	

The Blank Line Buffer

To ensure the parser correctly identifies boundaries, you must leave one blank line before and after Headings (`##`), Horizontal Rules (`---`), and HTML Blocks (`<div>`).



Structural Hierarchy

Rule Alert: The space after the hash marks is mandatory. **##Heading** will fail to render; **## Heading** will succeed.

Syntax	Heading Level	Rendered Output
<code>## Heading</code>	H2	Large bold section heading
<code>### Heading</code>	H3	Medium bold subsection heading
<code>#### Heading</code>	H4	Smaller subheading (Rarely used)

Standard Typography & Links

Rule Alert: Never use three dashes at the start of the file unless opening YAML. Inside the body, it renders as a thin dividing line. No space between the] and (in hyperlinks.

`**Bold text**` for emphasis.

`*Italic text*` for titles or quotes.

`---`

`[fast.ai Lesson 3 Official Page]`
`(https://course.fast.ai/)`

Bold text for emphasis.

Italic text for titles or quotes.

[fast.ai Lesson 3 Official Page](https://course.fast.ai/)

Narrative Formatting: Paragraphs vs. Lists

To give visual space and control over styling, body content that reads like a list should be written as bold-prefix paragraphs. Reserve standard bullet lists strictly for reference sections.

Reserve for References

- Top-level item
 - Nested item
 - Doubly nested item

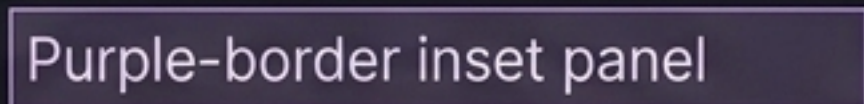
Preferred for Narrative

Personal productivity tools. Fine-tuned language models can automate mundane drafting tasks, allowing engineers to focus on architectural decisions.

Professional differentiation. In modern tech environments, understanding how to format and document your code is just as critical as writing it.

The Custom Layer: HTML & CSS Extensions

The most distinctive feature of the site is its use of custom `<div>` elements mapped to unique CSS classes.

Custom Class	Visual Appearance	Content Purpose	Dependencies
<code>.notebooklm-banner</code>	Dark navy gradient	Links	None
<code>.audio-section</code>	Dark block wrapping <code><audio></code>	Audio	None
<code>.guide-downloads</code>	Dark green box with buttons	Downloads	None
<code>.callout</code>	 Amber-border indented box	Important Concepts	None
<code>.swadia-point</code>	 Purple-border inset panel	Tutorial Points	None
<code>.testimonial</code>	 <i>Italic indented box with blue left border</i>	Quotes	Requires <code>.attribution</code>

Educational Callouts

Use **.callout** for core insights and **.swadia-point** for sequentially numbered tutorial key points. Notice the amber and purple visual accents.

The Callout

```
<div class="callout">
  <strong>Core insight:</strong> The loss collapses the entire
  dataset's
  prediction quality into a single number. Optimisation then becomes:
  decrease this number by adjusting the parameters.
</div>
```

Renders as: amber-border indented box with bold lead text.

The `.callout` CSS sets: background #1e1a0e, border-left: 4px solid #f9a825

Core insight: The loss collapses the entire dataset's prediction quality into a single number. Optimisation then becomes: decrease this number by adjusting the parameters.

The Swadia Point

```
<div class="swadia-point">
  <strong>Point 1 – Context First, Mechanics Second.</strong>
  Before any equation is introduced, establish why it matters...
</div>
```

Renders as: purple-border inset panel.

Used for the ten sequentially numbered tutorial key points.

1 Point 1 – Context First, Mechanics Second.
Before any equation is introduced, establish why it matters...

Complex Elements: The Testimonial Block

Some custom components require nested HTML tags. The `.testimonial` block relies on an inner `.attribution` div to correctly style the author line.

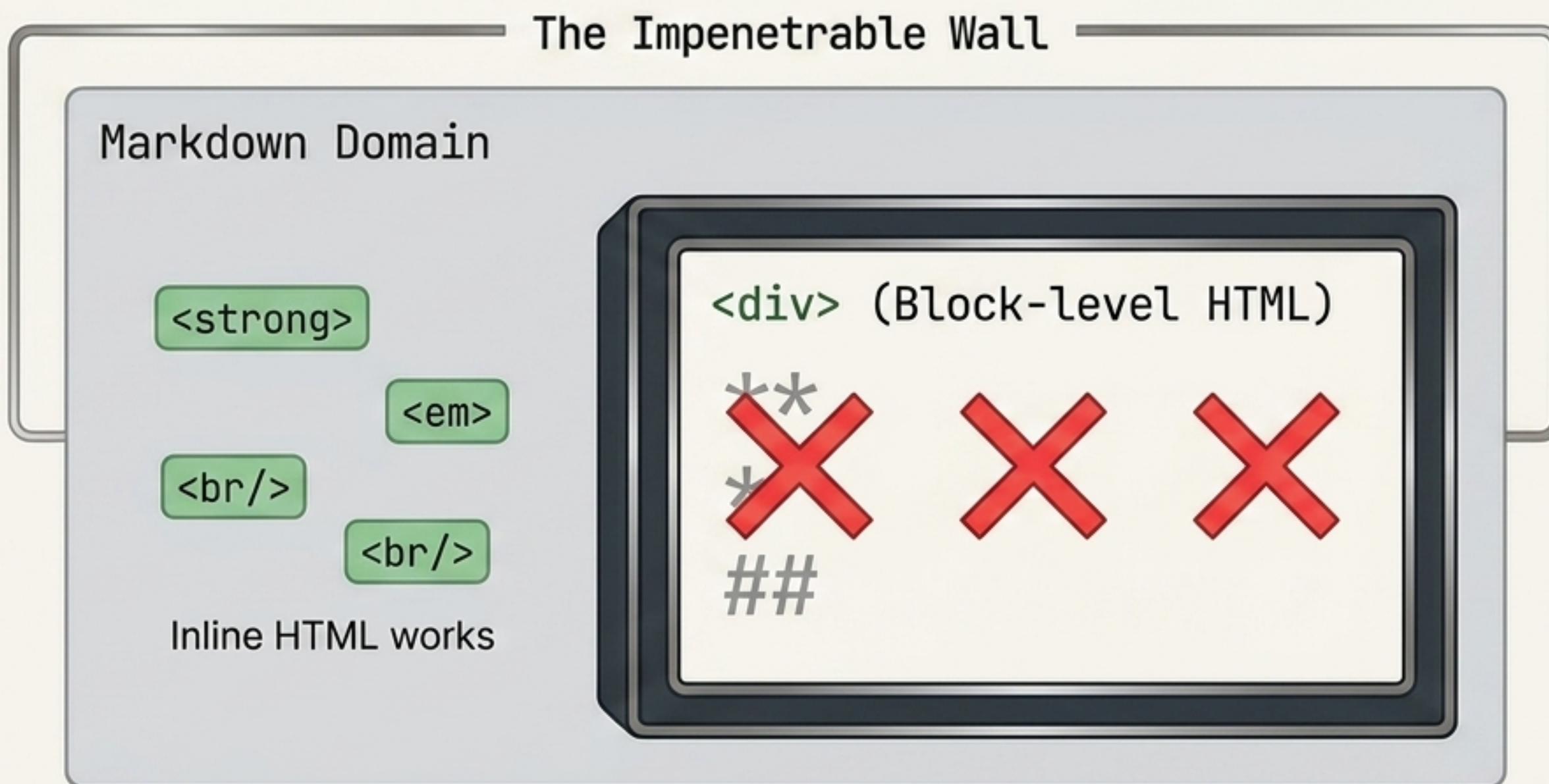
```
<div class="testimonial">  
  "I'd worked through three different  
  deep learning courses..."  
  
  <div class="attribution">  
    —Dr. M, Computational Biologist, UCL  
  </div>  
</div>
```

*"I'd worked through three different
deep learning courses..."*

—Dr. M, Computational Biologist, UCL

The Mixed-Syntax Guardrail

When you open a `<div>`, you cross an impenetrable wall. You must switch entirely to HTML (`<strong>` instead of `**bold*`) within the block.



Markdown vs. HTML in Practice

This is a Markdown heading

This is **Markdown** body text with a [link](https://example.com)(https://example.com).

Required Blank Line Buffer

```
<div class="callout">
```

Switch to HTML tags inside the block

```
  <strong>Inside HTML:</strong> Use HTML tags here, not Markdown.
```

```
  Two items: <em>item one</em> and <em>item two</em>.
```

```
</div>
```

Back to **Markdown** after the div.

The 9-Step Publication Checklist

Follow this sequence to ensure your post renders beautifully, routes correctly, and utilizes the full power of the platform.

